Wasi M.F. Naqvi

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Education

2026– **PhD Physics, McGill University** GPA: 4.00/4.00
Supervisor: Nicolas B. Cowan
Thesis: A Patchwork of Worlds: Artificial Intelligence for Population-Level Studies of Exoplanet Atmospheres

2025–26 **M.Sc. Physics, McGill University** GPA: 4.00/4.00
Supervisor: Nicolas B. Cowan
Thesis: HERMES: HiERarchical Modelling for Exoplanet Science

2021–25 **B.Sc. Physics (Honours), The University of British Columbia** GPA: 4.00/4.33
Minor in Data Science and Statistics
Thesis: Simulating In-Orbit Performance for the CASTOR Mission
Supervisor: Tyrone Woods
Distinctions: Dean's List, Best Thesis Award

Publications

Naqvi, W. M. F. and Cowan, N. B. ***HERMES: HiERarchical Modelling for Exoplanet Science.*** Royal Astronomical Society, Techniques and Instruments (2026,Submitted, under review.). Available on arxiv

Côté et al. (including **Naqvi, W.**). ***The CASTOR Mission***. Special Issue, Journal of Astronomical Telescopes, Instruments, and Systems, SPIE. (2025)

Awards & Honours

2025-	Kharusi Family International Science Fellowship, McGill University	$15,000 CAD
2021–25	Karen McKellin International Leader of Tomorrow Award, UBC	$250,000 CAD
2025	McGill Graduate Excellence Award	$5000 CAD

Contributed Talks & Posters

♁ Contributed Talk ♃ Poster

♁ Too Much Universe, Not Enough Grad Students: Deep Learning in Astrophysics — Aug 2026
Trottier Space Institute, McGill University — Montréal, Canada

♃ HERMES: HiERarchical Modelling for Exoplanet Science — Jun 2026
CASCA Annual General Meeting. — Montréal, Canada

♁ Comparative climatology of exoplanets using Hierarchical Modelling, — May 2026
Trottier Institute for Research on Exoplanets — Montréal, Canada

♁ **HERMES: HiERarchical Modelling for Exoplanet Science** — Mar 2026
Ariel Open Conference '26 — ESA Harwell, UK

♁ BayesBall: Bayesian Modelling for sport analytics, — Nov 2025

Trottier Space Institute Lunch Talks — Montréal, Canada

♁ A detector simulation pipeline for the CASTOR mission **(Invited)** — Jun 2026

The CASTOR Mission Consortium, NRC Herzberg Astronomy & Astrophysics — Victoria, Canada

Research & Work Experience

Trottier Institute for Research on Exoplanets, McGill University — Montréal, QC
Summer Researcher — May 2025–Present
Hierarchical Bayesian Modelling of planetary targets to probe population-level trends for the Ariel Space Mission.

NRC Herzberg Astronomy & Astrophysics — Victoria, BC
Junior Adaptive Optics Research Scientist — Jan 2025–Present
Implementing a convolutional neural network on an Adaptive Optics test bench for the 1.2 m REVOLT Telescope at the Dominion Astrophysical Observatory. Running machine-learning simulations alongside lab experiments under Dr. Maaike van Kooten.

TRIUMF — Vancouver, BC
Universal Polarization Research Assistant — Sep 2024–Present
Devising a Faraday Rotation method to measure polarization of optically-pumped Rb vapour at ISAC-II. Writing numerical simulations for polarizer experiments and developing a universal laser-polarization method for isotopes with unknown atomic transitions.

NRC Herzberg Astronomy & Astrophysics — Victoria, BC
Junior Research Scientist — May–Aug 2024
Designed Python software pipelines for imaging-sensor simulations and detector modelling for the CASTOR mission (Canadian Space Agency / NRC). Built with Dask (parallel computing) and ESA Pyxel (detector physics).

COCANA Lab, UBC Okanagan — Kelowna, BC
Undergraduate Research Assistant — May–Aug 2023
Optimization research for a commercial road-building project (SoftTree Inc.) under Dr. Yves Lucet. Used YALMIP solver; translated codebases between MATLAB and Python with unit-test verification.

CAN-SBX Stratospheric Balloon Design Team — Kelowna, BC
Physics & Data Science Lead — Sep 2023–May 2024
Led physics instrumentation and data analysis. Developed efficient MicroPython / C++ firmware for flight micro-controllers, designed ML models for atmospheric prediction, and assisted with dual-polarising antenna calibration and circuit analysis.

Teaching Assistantships

PHYS 180 Space, Time and Matter — Sep-Dec 2025
PHYS 241 Signal Processing — Jan-Apr 2025

Technical Competitions

Primary role across competitions: building, packaging, and benchmarking Bayesian models, U-Nets, Transformers, and CNNs for scientific data challenges.

ExoHack: Ariel Machine Learning Hackathon (**Winner**) — Mar 2026
NeurIPS Ariel Data Challenge — Jun 2026
Rocky Worlds DDT Challenge — May 2026

Community Engagement & Outreach

Founding Member & Organizing Committee, Montréal AstroStats Day — 2026–

Author, Astrobites	2025–
Mentor & Teaching Assistant, Small Private Online Courses, Scientists for Displaced People	2026–
iREx EDI Committee Member	2025–
McGill Hackathon Mentor	2026–
Astro on Tap Volunteer	2026–
Volunteer, McGill Physics “World of STEM” high-school astronomy sessions	2025–
Coordinator, graduate student lunches with TSI seminar speaker	2025–
Facilitator, McGill “From Planets to Particles” family science outreach fair	2025–
Organizing & Administration Volunteer, Exoclimes 2025 Conference	2025-
Science Outreach, Trottier Institute for Research on Exoplanets	2025–
UBCO Physics Student Club Treasurer	2021–2025
Exercise is Medicine Coordinator	2023–2025
Volunteer Soccer Coach, Royal Oak, Victoria	2024–2025
Mentor, CERN BeamLine For Schools (BL4S), Cedar College, Karachi	2021–
Science Outreach, Let’s Talk Science Okanagan	2023–2024
Volunteer, Okanagan Food Bank	2022–2024
Volunteer, Student Union Okanagan, UBC	2022–2023

Skills

Programming: Python, C, C++, Julia, Fortran, JavaScript, R, SQL, Shell/Bash
Markup & Typesetting: LaTeX, HTML/CSS, Markdown
Python Ecosystem: PyTorch, JAX, NumPyro, PyMC, Emcee, NumPy, SciPy, Astropy, Scikit-learn, Dask, Matplotlib, Pandas, ArviZ, Corner, Dynesty
Statistical & ML Methods: Hierarchical Bayesian Modelling, Gaussian Processes, MCMC Sampling, Nested Sampling, Simulation-Based Inference, PCA, CNNs, U-Nets, Transformers, Variational Inference, Model Comparison and Bench Marking
N-Body Codes: REBOUND
Computing Infrastructure: CANFAR, DRAC Supercomputing Clusters, Slurm, MPI, Git/GitHub, Docker, Linux
Other Software: Microsoft Excel, Google Sheets

Languages

Fluent: English, Urdu, Farsi/Dari (Persian)
Conversational: Hindi, Punjabi
Basic: French, Arabic

Relevant Coursework

Radiative Processes in Astrophysics, Statistical Learning, Computational Astrophysics, Machine Learning Theory, Probability Theory, Stochastic Processes, Exoplanets and Brown Dwarfs.